

Dian Xie

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Educational Background

Hefei University of Technology

Hefei, P.R.China

B.Eng. in Electronic Information Science and Technology

Sep. 2019 – Jun. 2023

Ranking: 1/147 GPA: 3.77/4.3 Supervisors: Prof. Shijie Hao, Prof. Chunhua Li

Publications

BrainRAM: Cross-Modality Retrieval-Augmented Image Reconstruction from Human Brain Activity

Dian Xie, Peiang Zhao, Jiarui Zhang, Kangqi Wei, Xiaobao Ni, Jiong Xia

ACM Multimedia (ACM MM), 2024. Selected as an Oral presentation paper (Top 3.97%).

A Lightness-aware Loss for Low-light Image Enhancement

Dian Xie, Huajun Xing, Liangyu Chen, Shijie Hao

Pattern Recognition Letters, 2023.

The Design of Multifunctional Intelligent Crutch System

Shilong Liu, **Dian Xie**, Zhiyuan Tang, Hao Wu, Lianlian Hua, Chunhua Li

Internet of Things Technologies, 2024.

A Speech Interaction Method for Intelligent Guided Crutch

Shilong Liu, **Dian Xie**, Zhiyuan Tang, Hao Wu, Lianlian Hua, Chunhua Li

Internet of Things Technologies, 2024.

An Indoor Grid Trajectory Partitioning Guidance Device

Shilong Liu, **Dian Xie**, Zhiyuan Tang, Chunhua Li

Chinese Patent, CN219798338U, 2023.

Project and Research Experiences

Natural Image Reconstruction from Human Brain Activity

Oct. 2023 – Apr. 2024

- The paper was accepted by ACM Multimedia 2024 (CCF-A), and selected as an oral presentation (<3.97%).
- Use Generative models to reconstruct what people see from their brain activity.
- Inspired by the working mechanism of human memory, we use retrieval-augmented generation (RAG) to enhance the reconstruction fidelity.

A Lightness-aware Loss for Low-light Image Enhancement

Oct. 2021 – Jan. 2023

- The paper has been published on Pattern Recognition Letters (CCF-C).
- Devised a loss function to alleviate over-exposure in current low-light image enhancement models.
- Our loss is zero-referenced, which means it didn't require additional training data, and could be easily transferred to other models.

College Competition of Smart Car, Director

Mar. 2021 – Jul. 2022

- Received a sponsorship of 6000 yuan from the school. And received second prize of National College Competition of Smart Car in Anhui province.

- The team independently designed and assembled an intelligent car, pass through the track elements such as roundabouts and intersections, and complete the task of image and label recognition.
- Mainly responsible for the preparation of camera tracking algorithm, using neural network to identify images and labels. Also drawing part of PCB.

Intelligent blind stick based on multimodal learning, Core member May. 2021 – Apr. 2023

- As a National College Student Innovation Training Program, received a sponsorship of 20000 yuan.
- Built a smart blind stick that can map indoor scenes built based on Jetson Nano and LiDAR, with autonomous navigation and voice assistance system.
- Published 2 papers and applied 1 patent.

Wireless communication and supervision system based on CC3200, Director Nov. 2022

- Received third prize of National College Competition of Electronic Design
- Built a smart blind stick that can map indoor scenes built based on Jetson Nano and LiDAR, with autonomous navigation and voice assistance system.

Received Awards

Outstanding Graduates of Anhui Province	2023
National Scholarship	2022
National Scholarship	2021
National Scholarship	2020
First Prize Anhui Robotic Competition	2020
Second Prize NCCCU Programming Contest (C++)	2020
Third Prize NCCCU Contest in Embedded System	2021
Second Prize in Anhui National College Competition of Smart Car	2022
Third Prize in Anhui National College Competition of Electronic Design	2022
Third Prize in Anhui National College Mathematical Contest in Modeling	2021
Third Prize Anhui Big Data and Artificial Intelligence Competition	2021

Services

Lab of Multimedia Computing, HFUT Hefei, P.R.China
 Undergraduate Research Intern Sep. 2021 – Apr. 2024
 Responsible for research in the fields of computational photography and neural decoding.

Electronic Design Innovation Lab, HFUT Hefei, P.R.China
 Student Director Oct. 2020 – Jun. 2023
 Participate in electronic-related competitions. Also responsible for organizing the training and recruiting procedure of fresh members.

Skills

Language	Mandarin Chinese (Native), English (Fluent, CET6: 566)
Programming	Python, C++, Java, Matlab
Development tools	Pytorch, LaTeX, Altium Designer, Keil MDK
Professional certificate	Computer Examination Band II (C++, Good)